TIPA Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	**Sensor-less Real-time 3D Sports Motion Tracking via Dual-Stage Depth Estimation on Single-View Broadcasting Video**			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI (√), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1)]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Knowledge Services		*Information & Communication*	
	Medium Category	Small Category	Medium Category	Small Category
	Added Value	*Broadcasting Media-Related Technologies*	*Digital Content*	*Digital Content Production and Distribution*
Concept of Proposed Technology	**Core Concept** This technology is an autonomous, non-intrusive 3D sports motion analytics pipeline designed to overcome the fundamental limitations of conventional wearable-based or multi-view tracking systems, which often require complex hardware installations, athlete cooperation, and expensive infrastructure. While various computer vision-based tracking methods have been proposed, existing frameworks (e.g., YOLO-based or general segmentation models) frequently suffer from depth flickering and spatial inconsistency, failing to maintain accurate 3D trajectories during complex athletic maneuvers. (1-3) The proposed platform integrates a dual-stage depth estimation architecture to ensure physical consistency in unconstrained broadcasting environments. The first stage, global localization (arena geometry-based), utilizes a proprietary court-line segmentation model to establish a stable physical reference, mitigating the errors induced by player occlusions and camera perspective shifts. The			

second stage, local localization (body mechanics-based), performs high-fidelity 3D pose lifting to extract joint-level depth information. This hierarchical approach addresses the depth ambiguity challenge in single-view videos, which is a well-documented bottleneck in sports biomechanics. (4-5)

Furthermore, the pipeline is optimized for low-latency, real-time processing (over 30 FPS on standard cloud GPU instances like g4dn), enabling immediate broadcasting overlays and coaching feedback. By decoupling 3D motion capture from physical sensors, this technology facilitates a scalable data-driven ecosystem for youth and amateur sports where high-end laboratory equipment is inaccessible. (1,5)

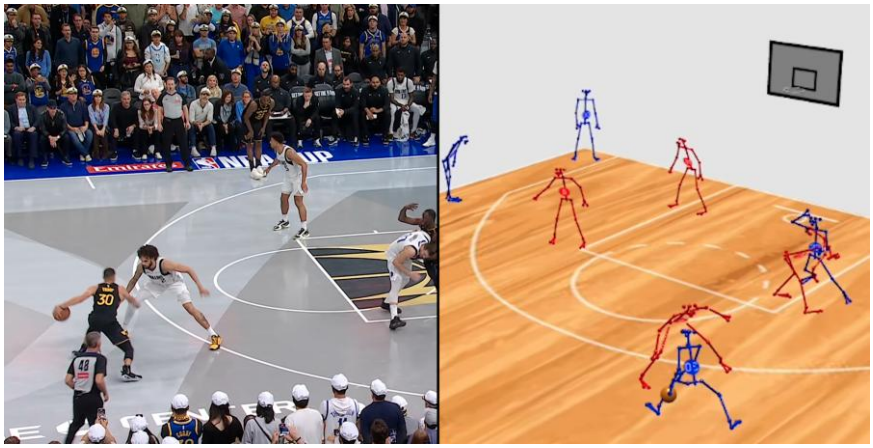

Figure: NBA Player Data Vision Tracking System, which requires multiple high-cost capture devices. This research aims to enable comparable tracking technology with a single low-cost camera.

Key Functions and Potential Applications

- Physics-Grounded Global Localization (Global Depth): Automatically maps players to 3D arena coordinates with sub-15cm accuracy by leveraging court-line geometry, ensuring stable tracking even during occlusions or aerial movements. (1-2)
- Spatio-Temporal Local Localization (Local Depth): Restores 3D skeletal kinematics by fusing 2D pose data with a deep-learning-based lifting model, maintaining temporal consistency across frames to prevent depth-jittering. (3-4)
- Cloud-to-Edge Optimized Pipeline: Achieves real-time inference on commercial GPU servers (e.g., AWS *g4dn*) with a modular architecture that supports future deployment on edge devices for on-site coaching. (5)
- Automatic Directing & Metadata Synchronization: Generates professional-grade highlight reels by controlling virtual camerawork (zoom/pan) based on 3D spatial data and synchronizing video with play-by-play records. (1-3)
- Application Areas: Elite scouting and recruitment, youth sports performance analytics, automated broadcasting for amateur leagues, and interactive fan engagement platforms.

Rationale for Joint R&D Project

Critical Technical Barriers

Conventional sports analytics and 3D motion capture technologies typically rely on wearable IMU sensors or synchronized multi-camera arrays to ensure spatial accuracy. However, they inherently

	suffer from intrinsic physical constraints in actual match environments where attaching hardware to athletes is strictly prohibited or logistically impossible. While computer vision-based monocular 3D pose estimation has emerged as a non-intrusive alternative, current models frequently struggle with depth ambiguity and spatio-temporal inconsistency. These issues—where high-speed maneuvers and player-to-player occlusions lead to significant flickering and coordinate drift—prevent the extraction of professional-grade biomechanical data from standard broadcasting feeds. (1-2, 5) The inability to maintain a stable global reference frame without physical markers remains a significant hurdle for field-scale deployment in the global youth and amateur sports sectors. (3) **Why Global Collaboration?** This joint R&D initiative seeks to structurally overcome these barriers by integrating advanced geometric deep learning with large-scale domain-specific data, moving beyond the limitations of standalone domestic research. ● Advancing Spatio-Temporal Consistency (Overseas Research Institute): By leveraging the overseas partner's expertise in geometric deep learning and temporal modeling, the project will develop a sophisticated 3D lifting architecture. This model will incorporate human-body kinematics as a physical prior to resolve the flickering issues inherent in monocular depth estimation. Such a physics-informed learning paradigm is established as the most effective approach for achieving high-fidelity motion reconstruction in crowded, dynamic environments. (4-5) ● Physics-Grounded Global Localization (Korean SME): The Korean SME provides a proprietary, large-scale sports dataset and high-precision arena geometry-based localization technology. This domestic expertise in court-line recognition and spatial mapping allows for the establishment of a global depth reference. By integrating this with the overseas partner's local depth estimation, the project will achieve a dual-stage depth estimation framework, preempting a global standard for sensor-less 3D tracking that domestic datasets alone cannot sustain due to the diversity of North American sports infrastructures. (1-2, 4) Through this strategic international partnership, the project will deliver: ● Technological Sovereignty: Securing a super-gap 3D tracking technology that functions solely on 2D video feeds, effectively replacing expensive hardware-dependent systems. ● Cost-Disruptive Democratization: Transforming the 3D motion capture market from a high-barrier, infrastructure-heavy model into a scalable, general video analysis market, specifically targeting the 100,000+ travel teams in the U.S. (1, 5) ● Global Scalability and Credibility: Establishing international-level performance benchmarks verified in North American environments, thereby ensuring the global applicability of the SaaS-based platform and creating high entry barriers against latecomers in the AI sports-tech market. (2-3)
Key Words	1. Sensor-less 3D Tracking 2. Computer Vision 3. Global/Local Localization

	4. Real-time Pose Estimation 5. Geometric Deep Learning

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (3)	End of the Project	Level (7)

R&D Objectives

R&D Objectives

To enable high-precision, sensor-less 3D motion tracking and biomechanical analysis of athletes using only single-view 2D broadcasting video, this project aims to develop, validate, and establish a commercialization basis for a universal 3D sports analytics platform that integrates:

- a physics-grounded arena geometry mapping engine for absolute spatial positioning,
- a dual-stage deep learning-based depth estimation (global & local) algorithm to ensure spatio-temporal kinematic consistency, and
- a low-latency real-time processing pipeline optimized for cloud-to-edge deployment, thereby eliminating the need for wearable sensors and dedicated multi-camera infrastructure.

Detailed Objectives

1. **Arena Geometry-based Global Localization**
   - Develop an advanced arena line recognition and segmentation model that establishes a robust 3D physical reference frame even in complex, occluded environments.
   - Achieve high-precision global depth estimation, specifying the absolute 3D coordinates of multiple players within the arena with a mean error of less than 15cm, serving as the spatial foundation for non-intrusive tracking.
2. **High-Fidelity 3D Pose Lifting & Local Localization**
   - Develop a local depth estimation algorithm that lifts 2D joint data into 3D space by fusing the specific global position with human body kinematics.
   - Implement a temporal consistency framework to restore seamless 3D skeletal structures, effectively mitigating the depth flickering issue and ensuring physically plausible motion trajectories during high-speed athletic actions.
3. **Real-Time Optimization & Scalable Deployment**
   - Optimize the entire end-to-end pipeline to achieve real-time inference (over 30 FPS) on commercial cloud GPU instances (e.g., AWS g4dn), enabling live broadcasting integration.
   - Establish a scalable deployment strategy for edge devices (mobile/on-site AI) through model pruning and quantization, ensuring the technology's applicability in diverse field environments ranging from elite stadiums to youth sports facilities.

R&D Target Specifications

Key Performance Indicator	Target Value
Global Localization Error	≤ 15cm (Absolute position within the arena using line geometry reference)
Local 3D Pose Accuracy	MPJPE ≤ 50mm (World-class level for non-sensor-based)
Depth Consistency	< 3% deviation between continuous frames (Including motion blur and occlusion)
Processing Speed (Real-time)	≥ 30 FPS (on g4dn, end-to-end pipeline basis)

R&D Plan	**Year 1: Global Localization and Arena Geometry Foundations** Research Institute ● Geometric Modeling of Arena Environments: Analyze the mathematical relationship between varying camera perspectives and arena line geometries to identify sources of spatial projection uncertainty. ● Advanced Semantic Line Recognition: Design an initial deep-learning framework for the semantic segmentation of complex arena lines, moving beyond traditional bounding-box methods to support robust homography under diverse lighting and occlusions. ● Standardization of Global Depth Protocols: Establish data acquisition and labeling protocols for various sports infrastructures (e.g., NCAA basketball courts, multi-purpose arenas) to ensure global model applicability. Korean SME Partner ● Global Localization Engine Development: Implement a baseline algorithm to specify the player's 3D position by correlating recognized arena lines with player detection points and build a first-stage spatial mapping prototype. ● Large-scale Sports Dataset Curation: Construct a domestic pilot dataset focused on diverse arena line configurations and player movement patterns to train the initial semantic segmentation model. ● Physics-Grounded Reference Setup: Develop a proprietary arena-reference coordinate system and conduct initial accuracy tests to verify the sub-15cm error margin in static environments. **Year 2: 3D Joint Restoration and Spatiotemporal Consistency** Overseas Research Institute ● 3D Pose Lifting Architecture Design: Develop a biomechanically informed lifting model that transforms 2D pose data into 3D coordinates by integrating spatial constraints and human-body kinematics. ● Temporal Consistency Modeling: Design spatiotemporal smoothing algorithms (e.g., Transformer-based sequential models) to resolve depth flickering and maintain seamless 3D skeletons during rapid motions. ● Overseas Data Validation & Benchmarking: Support the collection of multi-view validation datasets from North American sports facilities and establish an internationally comparable benchmarking framework for 3D motion accuracy. Korean SME Partner ● Integrated Local Localization Pipeline: Incorporate the 3D lifting model into the SME's existing pipeline and implement a transfer-learning module for rapid adaptation across different sports categories. ● Cloud-based Analysis Environment Setup: Deploy the Rev. B prototype on cloud instances (e.g., AWS g4dn) and perform initial performance tuning for multi-player 3D tracking. ● Metadata Synchronization Interface: Develop an MVP for a data management system that synchronizes 3D skeletal data with play-by-play game records for automated highlight generation.

	Year 3: Real-time System Optimization and Global Validation Overseas Research Institute ● Biomechanical Reliability Validation: Conduct independent analysis of 3D motion tracking performance under real-world match conditions and provide scientific validation of the system's robustness. ● Finalization of International Protocols: Finalize analysis protocols and evaluation methodologies for global SaaS deployment and support technical documentation for international certifications. ● Advanced Kinematic Feedback Logic: Provide expert guidance on extracting high-value coaching insights (e.g., offensive & defensive motion analysis) from the finalized 3D skeletal trajectories. Korean SME Partner ● Real-time Computational Optimization: Optimize the entire end-to-end pipeline using TensorRT and other acceleration frameworks to achieve stable real-time inference (over 30 FPS) on commercial GPU servers. ● Edge-Device Deployment & Lightweighting: Execute model pruning and quantization to verify the feasibility of on-device AI processing for mobile and low-spec edge devices (stretch target). ● Global SaaS Commercialization: Finalize product requirements through field demonstrations in both domestic and North American markets, secure official test reports from accredited bodies, and launch the commercial platform.

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	Advancing High-Fidelity 3D Motion Reconstruction and Spatiotemporal Optimization for Single-View Sports Analytics ● Design and enhance state-of-the-art 3D lifting neural network architectures to mathematically resolve the inherent depth ambiguity in single-view broadcasting videos, leveraging biomechanical constraints to ensure physically plausible motion reconstruction. ● Develop Transformer-based sequential modeling and spatiotemporal context optimization algorithms to maintain strict data consistency between frames and effectively mitigate depth flickering during high-speed athletic maneuvers. ● Research and implement lightweight, high-efficiency algorithms optimized for real-time inference on commercial GPU instances and edge devices, focusing on reducing computational redundancy while preserving tracking precision. ● Establish standardized protocols for 3D pose data acquisition and labeling across diverse international sports environments to minimize domain shift and ensure the universal generalization of the proposed analytics platform. ● Conduct independent performance benchmarking and scientific verification through overseas testbeds to produce validated data packages and evaluation reports that guarantee the global credibility of the 3D motion tracking system.
Expected R&R of Korean SME Partner	Leading the development of a physical reference engine and commercialization of an integrated sports analytics platform ● Develop and institutionalize a proprietary physical reference engine and spatial mapping database, leveraging large-scale arena line training data to ensure sub-15cm global localization accuracy; optimize the engine for mass-production readiness, focusing on reliability and stability across various sports court configurations. ● Develop and operate a cloud-connected SaaS platform and user-oriented analytics dashboard, enabling real-time metadata synchronization, AI model version management, and automated performance logging for diverse sports categories. ● Lead domestic and international field testing and performance tuning, integrating researched AI models into existing broadcasting and CCTV infrastructure to verify the end-to-end reliability of automated highlight generation and 3D data-driven reporting services. ● Execute global commercialization and market expansion strategies, including obtaining regional certifications (e.g., safety and data privacy approvals), securing accredited test reports for AI performance, and establishing sales/partnership networks in the North American and East Asian sports-tech markets.

5. Expected Outcomes/Impact

Expected Outcomes and Impact	● Technical Outcomes: Establishing a globally standardized 3D sports motion analytics framework ■ Secure source technology for sensor-less 3D motion tracking and a real-time cloud/edge-optimized pipeline (over 30 FPS) capable of converting single-view broadcasting videos into high-fidelity kinematic data without wearable sensors. ■ Build a global 3D sports motion database covering diverse athletic environments and disciplines, including: synchronized 2D-3D skeletal metadata, arena geometry-based physical references, and player-specific ID logs. ■ Establish and validate standardized protocols for 3D kinematic accuracy and benchmarking/evaluation methodologies, supported by international joint field-test results and scientific verification reports from overseas testbeds. ■ File core technology patents (domestic and international, including PCT) and secure accredited AI performance test reports and country-specific certifications (e.g., data privacy and safety) to establish high global entry barriers. ● Economic/Industrial Impact: Commercialization of a scalable SaaS platform and global market preemption ■ Preempt the global youth and amateur sports market by exporting a subscription-based (SaaS) 3D coaching solution, targeting 100,000+ travel teams in the U.S. and Europe without requiring expensive hardware infrastructure. ■ Revolutionize AI-driven sports broadcasting by automating professional-grade camera work (e.g., autonomous zoom/pan) and enabling high-value virtual ad overlays based on real-time spatial metadata, creating new revenue streams for lower-tier leagues. ■ Expand the sports data ecosystem through the commercialization of player lifecycle growth reports and AI-based scouting data models, facilitating a data-driven recruitment market for professional and collegiate sports. ■ Capture a significant share of the global AI sports analytics market—projected to reach USD 60.62 billion by 2035 with a 29.0% CAGR—by leveraging an 80% reduction in data analysis time through automation to address the high demand for performance and tactical analytics, which currently dominates 37.2% of the software-led market. ● Societal Impact: Democratization of elite performance data and fairness in talent discovery ■ Democratize elite-level performance analytics for youth and amateur athletes, providing equitable access to high-precision 3D data and coaching insights that were previously reserved for professional organizations with expensive sensor-based facilities. ■ Enhance transparency and objectivity in talent scouting through standardized performance metrics, reducing regional or economic biases and promoting fair opportunities for global sports talent discovery.

6. References

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